

Parth Katlana

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SUMMARY

Data Scientist with 3 years of experience in extracting actionable insights from large datasets. Proficient in Python, R, and SQL, with a strong foundation in machine learning, statistical analysis, and data visualization. Improved predictive models and optimize business strategies, leading to a 15% increase in customer retention and a 20% reduction in operational costs. communicating complex data findings to both technical and non-technical stakeholders. Leveraging data to drive decision-making and achieve business goals.

SKILLS

Languages: Python, R, SQL, SAS, C++, C, Java, JavaScript, React, HTML
IDEs and Methodologies: SDLC, Agile, Waterfall, Visual Studio Code, PyCharm, Jupyter Notebook
Machine Learning Algorithms: Linear Regression, Logistic Regression, Supervised Learning, Unsupervised Learning, Classification techniques, SVM, Random Forests, Naive Bayes, KNN, Deep Learning, Artificial Intelligence, Neural Networks, CNN, Gradient Boosting, XGBoost, RNN, LSTM, Quantum Machine Learning, Large-Language Models, Bagging, Image Processing, ETL, Neural Networks
Libraries: NumPy, Pandas, SciPy, Seaborn, TensorFlow, PyTorch, Ggplot2, Keras, Sklearn, NLTK, BeautifulSoup, OpenCV, HuggingFace
Visualization and Big Data Tools: Tableau, Power BI, GGplot, Google Power Query, Hadoop, Spark, Hive, Kafka, SAP
Databases and Containerization tools: MySQL, PostgreSQL, SQL Server, Oracle, MongoDB, Git, Docker, Kubernetes
Operating Systems and Cloud Platforms: MacOS, Windows, Linux, AWS, Azure, Google Cloud Platform, DataBricks
Artificial Intelligence Concepts: Generative AI, Natural Language Processing, Reinforcement Learning, Transfer Learning, Deep Learning
Data Analysis Analytics Tools: Geospatial, Time Series, Statistical and Exploratory Data Analysis, Advanced Statistical Models, D3.js, Bayesian Optimization, Gradient Descent, Google Analytics, Web Analytics, Adobe Analytics

EDUCATION

Master of Computer Science North Carolina State University, Raleigh, NC	Aug 2022 – May 2024
Bachelor of Engineering, Computer Science Symbiosis University, Indore, India	Aug 2018 – Jun 2022

EXPERIENCE

Berkshire Hathway, USA Jan 2024 – Present	Data Scientist
- Utilized SDLC, Agile, and Waterfall methodologies to develop and deploy ML models, resulting in streamlined project workflows and timely deliverables. - Architected scalable microservices using Docker and Kubernetes, improving scalability and reducing downtime. - Deployed machine learning models on AWS and Azure cloud platforms, ensuring high availability and scalability for pharma analytics applications. - Implemented machine learning models, including Random Forest and Logistic Regression, to predict patient outcomes, resulting in improvement in treatment accuracy. - Leveraged NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, and PyTorch packages for data manipulation, visualization, and model development and optimization, leading to a 30% reduction in data processing time. - Created interactive visualizations and reports using Tableau, Power BI, GGplot, and Google Power Query to communicate insights and findings to stakeholders, enhancing stakeholder understanding and engagement by 25%. - Managed databases including SQL Server, MySQL, PostgreSQL, and MongoDB for data storage and retrieval, improving data access and retrieval efficiency by 40%. Efficiently managed cloud platforms such as AWS, Azure, and GCP for data processing and storage. - Refined BERT models, optimizing data linkage across 1000+ product descriptions, resulting in a 30% uplift in data coherence. - Executed data pre-processing for efficient tokenization across diverse datasets, enhancing the analysis process by 20%, also achieved a 91% accuracy in distilling insights from complex datasets, underpinning key strategic decisions. - Implemented scalable infrastructure solutions, reducing model deployment time by 40% and enabling faster experimentation cycles.	
Laboratory of Analytical Sciences, USA May 2023 – Dec 2023	Graduate Research Assitant
Developed a reference architecture by utilizing Minstral, Claude, Llama-2, GPT-4, Phi, Falcon LLMs to measure ensemble behavior and perform text summarization on whitepaper, Fannie Mae documents (MapReduce, Refine). Fine-tuning and domain adaptation across temperature and prompts; implemented in AWS Jumpstar.	

- Pioneered an algorithm to measure Hallucinations on Abstractive/Hybrid summaries and evaluated their quality using different unsupervised summarization metrics. Applied Sentence-BERT (SBERT) for fine-tuning sentence embedding of each utterance. Worked as a teaching assistant for 6 months.
- Refactored Multimodal Datasets, using ElasticSearch to enhance retrieval by 15% for the 6 million data points.

Bytes Soft Solution, India | Jan 2020 – Mar 2022

Data Scientist

- Engineered a data analytics platform using Python and SQL, enabling the company to make data-driven decisions, leading to a 20% increase in operational efficiency.
- Analyzing large volumes of structured and unstructured data using Python, R, Tableau and ML library such as Scikit Learn with an emphasis on to evaluate data for anomaly detection, trend analysis, and data mining to support various business objectives.
- Conducted data analysis using Google Analytics, and Adobe Analytics to identify patterns and trends in user behaviour.
- Optimized cloud infrastructure on AWS, reducing hosting costs by 15% while maintaining system performance and reliability.
- Worked on different libraries related to Data science and Machine learning like Scikit - learn, OpenCV, NumPy, SciPy, Matplotlib, pandas, SQL etc.
- Proficient in managing entire data science project life cycle and actively involved in all the phases of project life cycle including data acquisition, data cleaning, data engineering, features scaling, features engineering, statistical modeling, testing and validation and data visualization.
- Developed and deployed machine learning models for Natural Language Processing (NLP), Reinforcement Learning, Transfer Learning, and Computer Vision applications, resulting in a 25% increase in accuracy for text classification tasks.
- Executed big data technologies including Hadoop, Spark, Hive, and Kafka for processing and analysing large volumes of data, leading to a 30% improvement in data processing speed and scalability.
- Devised a robust database architecture with MySQL and MongoDB, ensuring data integrity and enhancing query performance.
- Spearheaded the use of machine learning models in product recommendations, boosting sales conversion rates by 22%.
- Created interactive data visualization tools using Tableau and Power BI, facilitating real-time data analysis and improving business
- Collaborated with data engineers and software developers to integrate machine learning models into production systems, reducing deployment time by 20% and enhancing system performance.
- Facilitated to development of Generative AI models including GANs and VAEs for creative applications, resulting in the generation of high-quality synthetic data sets with a 35% reduction in training time.

PROJECTS

EyeReckon| Computer Vision, Video processing, Deep Learning, Neural Networks

Jan 2021 - Mar 2022

- Designed fine-tuning capability for the in-house CLIP model and YOLOv8 to enable vertical-specific model improvement resulting in 0.89 mAP50-95(B) for object detection.
- Improved model inferencing efficiency for models with Triton Inference Server, leveraging GPU acceleration and made available as APIs; reducing the deployment time by an average of 2 seconds by using key-value storage systems.
- Deployed a feature to production for dynamically rendering geospatial data on a map, orchestrating seamless API calls to query MongoDB, and efficiently triggering real-time visual updates through Vue.js components.

Integrated Learning Hub | Cloud Architecture, DynamoDB, CloudFront

Nov 2023 - Dec 2023

- Designed and implemented a scalable Integrated Learning Hub on AWS, capable of handling over 1 million simultaneous user logins and improving content delivery efficiency.
- Enhanced platform security by incorporating AWS Cognito for secure authentication and AWS WAF to guard against SQL injection attacks.
- Cut operational expenses by up to 75% by utilizing serverless architecture through AWS Lambda and Elastic Beanstalk for Service Resource Management (SRM).

PUBLICATIONS

- Parth K., amp; Grant Forbes “ Metric Ensemble for hallucination detection”. [Link to Paper](#)
- Zeydy Ortiz, amp; Parth K.“Uncovering Factual Consistency Errors”. [Link to Paper](#)
- Marley C. amp; Parth K. amp; Chris K. amp; Akhil N.“General Interface for Factored State Optimization Problems”. [Link to Paper](#)